”BASIC OF DIFFERENTIATION AND IT‘S APPLICATIONS“

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by

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CERTIFICATE

This is to certify that this project entitled, **"BASIC OF DIFFERENTIATION AND IT'S APPLICATIONS"** submitted by

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for the Degree of the **BACHELOR OF SCIENCE IN MATHEMATICS** to Madurai Kamaraj University is based on the results of the studies carried out by the student under my guidance and supervision. It is further certified that all principles of academic honesty and integrity as advocated by Madurai Kamaraj University are strictly adhered to.

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1 INTRODUCTION

What is Differentiation:

Differentiation is one of the fundamental concepts of calculus. It is mainly used to find the rate at which one quantity changes with respect to another quantity.

In simple words, differentiation helps us understand how fast something is changing.

For example, when a car moves, its position changes with time. Differentiation can be used to find the velocity of the car. Similarly, differentiating velocity with respect to time gives acceleration.

Differentiation is not limited to mathematics. It has many applications in physics, engineering, economics, statistics, biology, business and other fields.

Definition of Differentiation:

If $y = f(x)$, then the derivative of y with respect to x is written as:

$$\frac{dy}{dx}$$

It represents the instantaneous rate of change of y with respect to x.

The derivative can be defined using the limit:

$$\frac{dy}{dx} = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

Simple Example of Differentiation:

Consider the function

$$y = x^2$$

Differentiating both sides with respect to x,

$$\frac{dy}{dx} = 2x$$

If $x = 3$, then

$$\frac{dy}{dx} = 2(3)$$

Therefore,

$$\frac{dy}{dx} = 6$$

Hence, the rate of change of the function at $x = 3$ is

$$\boxed{6}$$

Differentiation in Motion:

Differentiation has an important application in studying the motion of an object.

Let s represent displacement and t represent time. The velocity of an object is obtained by differentiating displacement with respect to time.

$$v = \frac{ds}{dt}$$

Similarly, acceleration is obtained by differentiating velocity with respect to time.

$$a = \frac{dv}{dt}$$

Therefore,

$$a = \frac{d^2s}{dt^2}$$

Thus, differentiation provides a mathematical relationship between displacement, velocity and acceleration.

Importance of Differentiation:

Differentiation is important because it helps us to:

- Find the rate of change of quantities.
- Calculate velocity and acceleration.
- Find maximum and minimum values.
- Study the behaviour of functions.
- Solve optimization problems.
- Analyse cost, revenue and profit.
- Study population growth.
- Understand changes in physical and natural phenomena.

Applications of Differentiation:

Differentiation has many applications in different fields.

1. **Physics:**
   Differentiation is used to calculate velocity, acceleration and other rates of change in physical quantities.
2. **Economics:**
   Differentiation is used to study cost, revenue, profit and marginal quantities.
3. **Engineering:**
   Differentiation is used in optimization, design and analysis of engineering systems.
4. **Biology:**
   Differentiation can be used to study population growth and other biological changes.
5. **Business:**
   Differentiation helps businesses to maximize profit and minimize cost.
6. **Mathematics:**
   Differentiation is used to study functions, curves, maxima, minima and rates of change.

Aim of the Project:

The main aim of this project is to study the applications of differentiation in real-life situations and understand how derivatives are used to solve practical problems.

Objectives:

The objectives of this project are:

1. To understand the basic concept of differentiation.
2. To study different rules of differentiation.
3. To understand the concept of rate of change.
4. To apply differentiation to motion-related problems.
5. To study maxima and minima.
6. To understand applications in economics and business.
7. To demonstrate the importance of differentiation in real-life problems.

Scope of the Project:

The scope of this project is to understand how differentiation can be applied to different real-life situations. The project focuses on basic differentiation concepts and their applications in motion, optimization, economics, business and population studies.

The study shows that differentiation is not only a theoretical concept but also an important mathematical tool for solving practical problems.

2 BASIC DIFFERENTIATION RULES AND FORMULAS

Introduction

Differentiation is a basic concept in calculus used to find the rate of change of one quantity with respect to another. In real-life applications, differentiation helps us to study motion, growth, profit, cost, temperature, and many other changing quantities.

There are some basic rules and formulas which make differentiation easier and faster. These rules are used throughout calculus and are important for solving real-life problems.

Constant Rule

The derivative of a constant is zero.

If c is a constant, then

$$\frac{d}{dx}(c) = 0$$

Example:
Let

$$y = 8$$

Then,

$$\frac{dy}{dx} = 0$$

Therefore, the derivative of a constant is zero.

Power Rule

The power rule is one of the most commonly used rules in differentiation.

If

$$y = x^n$$

then

$$\frac{dy}{dx} = nx^{n-1}$$

where n is a constant.

Example:

Let

$$y = x^5$$

Using the power rule,

$$\frac{dy}{dx} = 5x^{5-1}$$

Therefore,

$$\frac{dy}{dx} = 5x^4$$

Constant Multiple Rule

If a function is multiplied by a constant, the constant remains unchanged during differentiation.

$$\frac{d}{dx}[cf(x)] = c\frac{d}{dx}[f(x)]$$

where c is a constant.

Example:

Let

$$y = 7x^3$$

Then,

$$\frac{dy}{dx} = 7\frac{d}{dx}(x^3)$$

Using the power rule,

$$\frac{dy}{dx} = 7(3x^2)$$

Therefore,

$$\frac{dy}{dx} = 21x^2$$

Sum Rule

The derivative of the sum of two or more functions is equal to the sum of their derivatives.

$$\frac{d}{dx}[f(x) + g(x)] = \frac{df}{dx} + \frac{dg}{dx}$$

Example:
Let

$$y = x^3 + x^2 + x$$

Then,

$$\frac{dy}{dx} = 3x^2 + 2x + 1$$

Therefore,

$$\boxed{\frac{dy}{dx} = 3x^2 + 2x + 1}$$

Difference Rule

The derivative of the difference of two functions is equal to the difference of their derivatives.

$$\frac{d}{dx}[f(x) - g(x)] = \frac{df}{dx} - \frac{dg}{dx}$$

Example:
Let

$$y = x^4 - x^2$$

Then,

$$\frac{dy}{dx} = 4x^3 - 2x$$

Therefore,

$$\boxed{\frac{dy}{dx} = 4x^3 - 2x}$$

Product Rule

When two functions are multiplied together, the product rule is used.
If

$$y = uv$$

then

$$\frac{dy}{dx} = u\frac{dv}{dx} + v\frac{du}{dx}$$

Example:
Let

$$y = x^2(x+3)$$

Here,

$$u = x^2$$

and

$$v = x + 3$$

Therefore,

$$\frac{du}{dx} = 2x$$

and

$$\frac{dv}{dx} = 1$$

Using the product rule,

$$\frac{dy}{dx} = x^2(1) + (x+3)(2x)$$

$$= x^2 + 2x^2 + 6x$$

Therefore,

$$\boxed{\frac{dy}{dx} = 3x^2 + 6x}$$

Quotient Rule

When one function is divided by another function, the quotient rule is used.

If

$$y = \frac{u}{v}$$

then

$$\frac{dy}{dx} = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$$

Example:

Let

$$y = \frac{x^2}{x+1}$$

Here,

$$u = x^2$$

and

$$v = x + 1$$

Therefore,

$$\frac{du}{dx} = 2x$$

and

$$\frac{dv}{dx} = 1$$

Using the quotient rule,

$$\frac{dy}{dx} = \frac{(x+1)(2x) - x^2(1)}{(x+1)^2}$$

Simplifying,

$$\frac{dy}{dx} = \frac{2x^2 + 2x - x^2}{(x+1)^2}$$

Therefore,

$$\boxed{\frac{dy}{dx} = \frac{x^2 + 2x}{(x+1)^2}}$$

Chain Rule

The chain rule is used when one function is inside another function.

If

$$y = f(g(x))$$

then

$$\frac{dy}{dx} = \frac{dy}{du}\frac{du}{dx}$$

where

$$u = g(x)$$

Example:

Let

$$y = (x^2 + 1)^3$$

Put

$$u = x^2 + 1$$

Then,

$$y = u^3$$

Therefore,

$$\frac{dy}{du} = 3u^2$$

and

$$\frac{du}{dx} = 2x$$

Using the chain rule,

$$\frac{dy}{dx} = 3u^2(2x)$$

Substituting $u = x^2 + 1$,

$$\boxed{\frac{dy}{dx} = 6x(x^2 + 1)^2}$$

Standard Differentiation Formulas

The following formulas are commonly used in differentiation.

- Constant:

$$\frac{d}{dx}(c) = 0$$

- Power function:

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

- Exponential function:

$$\frac{d}{dx}(e^x) = e^x$$

- Natural logarithmic function:

$$\frac{d}{dx}(\ln x) = \frac{1}{x}$$

- Sine function:

$$\frac{d}{dx}(\sin x) = \cos x$$

- Cosine function:

$$\frac{d}{dx}(\cos x) = -\sin x$$

- Tangent function:

$$\frac{d}{dx}(\tan x) = \sec^2 x$$

- Cotangent function:

$$\frac{d}{dx}(\cot x) = -\csc^2 x$$

- Secant function:

$$\frac{d}{dx}(\sec x) = \sec x \tan x$$

- Cosecant function:

$$\frac{d}{dx}(\csc x) = -\csc x \cot x$$

Worked Example 1

Find the derivative of

$$y = 5x^4$$

Using the power rule,

$$\frac{dy}{dx} = 5(4x^3)$$

Therefore,

$$\boxed{\frac{dy}{dx} = 20x^3}$$

Worked Example 2

Find the derivative of

$$y = 3x^3 + 5x^2 - 7x + 4$$

Differentiating each term,

$$\frac{dy}{dx} = 3(3x^2) + 5(2x) - 7 + 0$$

Therefore,

$$\boxed{\frac{dy}{dx} = 9x^2 + 10x - 7}$$

Worked Example 3

Find the derivative of

$$y = \sin x + x^2$$

Differentiating both terms,

$$\frac{dy}{dx} = \cos x + 2x$$

Therefore,

$$\boxed{\frac{dy}{dx} = \cos x + 2x}$$

Worked Example 4

Find the derivative of

$$y = e^x + \ln x$$

Using the standard formulas,

$$\frac{dy}{dx} = e^x + \frac{1}{x}$$

Therefore,

$$\boxed{\frac{dy}{dx} = e^x + \frac{1}{x}}$$

Worked Example 5

Find the derivative of

$$y = (3x + 2)^4$$

Using the chain rule,

$$\frac{dy}{dx} = 4(3x + 2)^3(3)$$

Therefore,

$$\boxed{\frac{dy}{dx} = 12(3x + 2)^3}$$

3 RATE OF CHANGE

Introduction

One of the most important applications of differentiation is finding the rate at which one quantity changes with respect to another quantity.

In real life, many quantities are continuously changing. For example, the position of a moving car changes with time, the temperature changes during the day, and the profit of a company changes with the number of products sold.

Differentiation provides a mathematical method to measure these changes.

Meaning of Rate of Change

Suppose y is a quantity that depends on another quantity x. The rate of change of y with respect to x is given by

$$\frac{dy}{dx}$$

This represents how much y changes when x changes by a small amount.

For example, if

$$y = x^2$$

then

$$\frac{dy}{dx} = 2x$$

Therefore, the rate of change of y depends on the value of x.

Average Rate of Change

The average rate of change of a function between two points is given by

$$\frac{\Delta y}{\Delta x} = \frac{y_2 - y_1}{x_2 - x_1}$$

It measures the average change in y for a given change in x.

Example:

Consider

$$y = x^2$$

Let

$$x_1 = 1, \qquad x_2 = 3$$

Then,

$$y_1 = 1^2 = 1$$

and

$$y_2 = 3^2 = 9$$

Therefore,

$$\frac{\Delta y}{\Delta x} = \frac{9 - 1}{3 - 1}$$

$$= \frac{8}{2}$$

Therefore,

$$\boxed{\frac{\Delta y}{\Delta x} = 4}$$

Hence, the average rate of change between $x = 1$ and $x = 3$ is 4.

Instantaneous Rate of Change

The instantaneous rate of change gives the rate of change at a particular point.

It is represented by the derivative

$$\frac{dy}{dx}$$

If

$$y = f(x)$$

then the instantaneous rate of change is

$$\frac{dy}{dx} = f'(x)$$

This concept is very useful in real-life applications such as speed, acceleration, population growth, and economic analysis.

Rate of Change in Motion

In motion, the position of an object changes with time.

Let the position of an object be

$$s = f(t)$$

where s is the distance and t is the time.

The velocity is the rate of change of distance with respect to time.

Therefore,

$$v = \frac{ds}{dt}$$

The acceleration is the rate of change of velocity with respect to time.

Therefore,

$$a = \frac{dv}{dt}$$

Since

$$v = \frac{ds}{dt}$$

we get

$$a = \frac{d^2s}{dt^2}$$

Thus, differentiation is directly used to find velocity and acceleration.

Worked Example – Velocity

Suppose the position of a car is given by

$$s = t^2 + 4t$$

where s is measured in metres and t is measured in seconds.

The velocity is

$$v = \frac{ds}{dt}$$

Differentiating,

$$v = 2t + 4$$

Therefore,

$$\boxed{v = 2t + 4}$$

At $t = 3$ seconds,

$$v = 2(3) + 4$$

$$v = 10$$

Therefore, the velocity of the car at 3 seconds is

$$\boxed{10 \text{ m/s}}$$

Worked Example – Acceleration

Consider the velocity of an object given by

$$v = 3t^2 + 2t$$

The acceleration is the rate of change of velocity.
Therefore,

$$a = \frac{dv}{dt}$$

Differentiating,

$$a = 6t + 2$$

At $t = 2$ seconds,

$$a = 6(2) + 2$$

$$a = 14$$

Therefore,

$$\boxed{a = 14 \text{ m/s}^2}$$

Rate of Change of Population

Differentiation can also be used to study population growth.
Suppose the population of a town is represented by

$$P(t) = 1000 + 50t + 2t^2$$

where t represents time.
The rate of population growth is

$$\frac{dP}{dt}$$

Therefore,

$$\frac{dP}{dt} = 50 + 4t$$

At $t = 5$,

$$\frac{dP}{dt} = 50 + 4(5)$$

$$= 70$$

Therefore, the population is increasing at a rate of

$$\boxed{70 \text{ people per unit time}}$$

Rate of Change in Temperature

Temperature changes continuously with time. Differentiation can be used to determine how quickly temperature changes.

Suppose the temperature is given by

$$T(t) = 20 + 3t - t^2$$

The rate of change of temperature is

$$\frac{dT}{dt}$$

Therefore,

$$\frac{dT}{dt} = 3 - 2t$$

At $t = 1$,

$$\frac{dT}{dt} = 3 - 2(1)$$

$$= 1$$

Therefore, at $t = 1$, the temperature is increasing at the rate of

$$\boxed{1^{\circ}\text{C per unit time}}$$

Rate of Change in Business

Differentiation is also useful in economics and business.

Let the cost of producing x units be

$$C(x) = 500 + 20x + 2x^2$$

The rate of change of cost with respect to the number of units is

$$\frac{dC}{dx}$$

Therefore,

$$\frac{dC}{dx} = 20 + 4x$$

This derivative represents the marginal cost of production.

Thus, differentiation helps businesses understand how the cost changes when the production level changes.

Importance of Rate of Change

The concept of rate of change is important in many areas.

- It helps to find the velocity of a moving object.
- It helps to calculate acceleration.
- It is used to study population growth.
- It helps to understand changes in temperature.
- It is used to calculate marginal cost and marginal revenue.
- It is useful in engineering and science.
- It helps to analyse changing quantities in real-life situations.

4 APPLICATION OF DIFFERENTIATION IN MOTION

Introduction

Differentiation plays an important role in studying the motion of objects. When an object moves, its position changes with time. By using differentiation, we can find the velocity and acceleration of the moving object.

The three important quantities used in the study of motion are:

- Distance or Position
- Velocity
- Acceleration

Differentiation provides a mathematical relationship between these quantities.

Position of a Moving Object

Let the position of an object at time t be represented by

$$s = f(t)$$

where s represents the position of the object and t represents time.
The position tells us where the object is located at a particular time.
For example, if

$$s = t^2 + 2t$$

then the position of the object depends on time.

Velocity

Velocity is defined as the rate of change of position with respect to time.

Therefore,

$$v = \frac{ds}{dt}$$

where

- v is velocity,
- s is position,
- t is time.

The unit of velocity is usually metres per second (m/s).

Acceleration

Acceleration is the rate of change of velocity with respect to time.

Therefore,

$$a = \frac{dv}{dt}$$

Since

$$v = \frac{ds}{dt}$$

we get

$$a = \frac{d^2s}{dt^2}$$

The unit of acceleration is usually metres per second squared (m/s^2).

Relationship Between Position, Velocity and Acceleration

The relationship between position, velocity, and acceleration can be written as

$$s \longrightarrow v \longrightarrow a$$

where

$$v = \frac{ds}{dt}$$

and

$$a = \frac{dv}{dt}$$

Therefore,

$$a = \frac{d^2s}{dt^2}$$

This shows that velocity is the first derivative of position and acceleration is the second derivative of position.

Worked Example 1 – Finding Velocity

Suppose the position of a particle is given by

$$s = t^3 + 2t^2 + 3t$$

Find the velocity at time $t = 2$ seconds.
The velocity is

$$v = \frac{ds}{dt}$$

Differentiating,

$$v = 3t^2 + 4t + 3$$

At $t = 2$,

$$v = 3(2)^2 + 4(2) + 3$$

$$= 12 + 8 + 3$$

Therefore,

$$\boxed{v = 23 \text{ m/s}}$$

Hence, the velocity of the particle at 2 seconds is 23 m/s.

Worked Example 2 – Finding Acceleration

Consider the position function

$$s = t^3 + 2t^2 + 3t$$

From the previous example, the velocity is

$$v = 3t^2 + 4t + 3$$

Acceleration is

$$a = \frac{dv}{dt}$$

Therefore,

$$a = 6t + 4$$

At $t = 2$,

$$a = 6(2) + 4$$

$$= 12 + 4$$

Therefore,

$$\boxed{a = 16 \text{ m/s}^2}$$

Hence, the acceleration of the particle at 2 seconds is 16 m/s^2.

Worked Example 3 – Finding Distance, Velocity and Acceleration

Let the position of a particle be

$$s = 5t^2 + 4t + 2$$

The velocity is

$$v = \frac{ds}{dt}$$

Therefore,

$$v = 10t + 4$$

The acceleration is

$$a = \frac{dv}{dt}$$

Therefore,

$$a = 10$$

Hence,

$$\boxed{v = 10t + 4}$$

and

$$\boxed{a = 10 \text{ m/s}^2}$$

Thus, the velocity changes with time, while the acceleration remains constant.

Motion of a Car

Consider a car moving along a straight road. Its position can be represented by a function of time.

Suppose

$$s = 2t^3 + 3t^2$$

The velocity of the car is

$$v = \frac{ds}{dt}$$

Therefore,

$$v = 6t^2 + 6t$$

The acceleration is

$$a = \frac{dv}{dt}$$

Therefore,

$$a = 12t + 6$$

At $t = 2$ seconds,

$$v = 6(2)^2 + 6(2)$$

$$= 24 + 12$$

$$\boxed{v = 36 \text{ m/s}}$$

The acceleration is

$$a = 12(2) + 6$$

$$\boxed{a = 30 \text{ m/s}^2}$$

Therefore, differentiation helps us determine both the velocity and acceleration of a moving car.

Motion Under Gravity

Differentiation can also be used to study an object moving vertically under gravity.

Suppose an object is thrown upward with initial velocity u. Its position after time t can be written as

$$s = ut - \frac{1}{2}gt^2$$

where g is the acceleration due to gravity.

Differentiating with respect to t,

$$v = \frac{ds}{dt}$$

Therefore,

$$v = u - gt$$

Differentiating again,

$$a = \frac{dv}{dt}$$

Hence,

$$\boxed{a = -g}$$

The negative sign indicates that acceleration is directed downward.

Worked Example 4 – Object Thrown Upward

Suppose an object is thrown upward with an initial velocity of

$$u = 20 \text{ m/s}$$

Take

$$g = 10 \text{ m/s}^2$$

The position is

$$s = 20t - 5t^2$$

The velocity is

$$v = \frac{ds}{dt}$$

Therefore,

$$v = 20 - 10t$$

At the highest point, the velocity becomes zero.
Therefore,

$$20 - 10t = 0$$

$$10t = 20$$

Hence,

$$\boxed{t = 2 \text{ seconds}}$$

Therefore, the object reaches its maximum height after 2 seconds.

Importance of Differentiation in Motion

Differentiation is widely used in the study of motion.

- It helps to find the velocity of a moving object.
- It helps to find acceleration.
- It helps to determine whether an object is speeding up or slowing down.
- It is used to study the motion of cars, trains, aircraft, and other vehicles.
- It is useful in physics and engineering.
- It helps to analyse objects moving under gravity.

Real-Life Applications

The application of differentiation in motion can be seen in many real-life situations.

1. **Transportation:** The speed and acceleration of vehicles can be analysed using derivatives.
2. **Sports:** The motion of a ball or an athlete can be studied using position, velocity, and acceleration.
3. **Engineering:** Engineers use differentiation to study the movement of machines and mechanical systems.
4. **Space Science:** The motion of satellites and spacecraft can be analysed using mathematical models.
5. **Physics:** Differentiation is fundamental for studying motion and forces.

5 MAXIMA AND MINIMA

Introduction

Differentiation is useful for finding the maximum and minimum values of a function. In real life, we often need to find the highest or lowest value of a quantity.

For example, a business may want to find the maximum profit, an engineer may want to minimize the cost of production, or a farmer may want to obtain the maximum area from a fixed amount of material.

The concepts of maxima and minima provide a mathematical method for solving such problems.

Meaning of Maximum and Minimum

A function has a maximum value at a point when the function changes from increasing to decreasing.

A function has a minimum value at a point when the function changes from decreasing to increasing.

These points are called turning points of the function.

Critical Point

A critical point occurs when the first derivative of a function is zero or does not exist.

For a differentiable function $y = f(x)$, we find critical points by solving

$$\frac{dy}{dx} = 0$$

These critical points are then tested to determine whether they represent a maximum or a minimum.

Condition for Maximum

Suppose

$$y = f(x)$$

At a critical point,

$$\frac{dy}{dx} = 0$$

If

$$\frac{d^2y}{dx^2} < 0$$

then the function has a maximum value at that point.

Therefore, the second derivative test for maximum is

$$\boxed{\frac{dy}{dx} = 0, \qquad \frac{d^2y}{dx^2} < 0}$$

Condition for Minimum

At a critical point,

$$\frac{dy}{dx} = 0$$

If

$$\frac{d^2y}{dx^2} > 0$$

then the function has a minimum value at that point.
Therefore, the second derivative test for minimum is

$$\boxed{\frac{dy}{dx} = 0, \qquad \frac{d^2y}{dx^2} > 0}$$

Worked Example – Finding a Maximum

Consider the function

$$y = -x^2 + 6x + 2$$

First, find the first derivative.

$$\frac{dy}{dx} = -2x + 6$$

At the critical point,

$$\frac{dy}{dx} = 0$$

Therefore,

$$-2x + 6 = 0$$

$$2x = 6$$

Hence,

$$x = 3$$

Now find the second derivative.

$$\frac{d^2y}{dx^2} = -2$$

Since

$$\frac{d^2y}{dx^2} < 0$$

the function has a maximum at $x = 3$.
The maximum value is

$$y = -(3)^2 + 6(3) + 2$$

$$= -9 + 18 + 2$$

Therefore,

$$\boxed{y = 11}$$

Hence, the maximum value of the function is 11.

Worked Example – Finding a Minimum

Consider the function

$$y = x^2 - 4x + 5$$

First, differentiate the function.

$$\frac{dy}{dx} = 2x - 4$$

At the critical point,

$$2x - 4 = 0$$

Therefore,

$$x = 2$$

Now find the second derivative.

$$\frac{d^2y}{dx^2} = 2$$

Since

$$\frac{d^2y}{dx^2} > 0$$

the function has a minimum at $x = 2$.
The minimum value is

$$y = (2)^2 - 4(2) + 5$$

$$= 4 - 8 + 5$$

Therefore,

$$\boxed{y = 1}$$

Hence, the minimum value of the function is 1.

Application in Finding Maximum Area

Differentiation can be used to find the maximum area of a rectangular object.
Suppose a rectangle has length x and width y, and its perimeter is fixed.
The perimeter is

$$2x + 2y = P$$

Therefore,

$$y = \frac{P}{2} - x$$

The area of the rectangle is

$$A = xy$$

Substituting the value of y,

$$A = x\left(\frac{P}{2} - x\right)$$

Therefore,

$$A = \frac{P}{2}x - x^2$$

Differentiate with respect to x.

$$\frac{dA}{dx} = \frac{P}{2} - 2x$$

For maximum area,

$$\frac{dA}{dx} = 0$$

Therefore,

$$\frac{P}{2} - 2x = 0$$

Hence,

$$x = \frac{P}{4}$$

Since

$$y = \frac{P}{2} - x$$

we get

$$y = \frac{P}{4}$$

Therefore, the maximum area occurs when

$$\boxed{x = y}$$

Thus, among all rectangles with a fixed perimeter, a square gives the maximum area.

Application in Business

Maxima and minima are widely used in business and economics.

A company may want to maximize its profit or minimize its production cost.

Let the profit function be

$$P(x) = R(x) - C(x)$$

where $R(x)$ represents revenue and $C(x)$ represents cost.

To find the maximum profit, differentiate the profit function.

$$\frac{dP}{dx} = 0$$

The critical point can then be tested using the second derivative.

If

$$\frac{d^2P}{dx^2} < 0$$

the profit is maximum.

Thus, differentiation helps businesses determine the most profitable level of production.

Application in Cost Minimization

A business may also want to reduce its production cost.

Suppose the cost function is

$$C(x) = x^2 - 10x + 50$$

The first derivative is

$$\frac{dC}{dx} = 2x - 10$$

For minimum cost,

$$2x - 10 = 0$$

Therefore,

$$x = 5$$

The second derivative is

$$\frac{d^2C}{dx^2} = 2$$

Since

$$\frac{d^2C}{dx^2} > 0$$

the cost is minimum at $x = 5$.

The minimum cost is

$$C(5) = 5^2 - 10(5) + 50$$

$$= 25 - 50 + 50$$

Therefore,

$$\boxed{C(5) = 25}$$

Real-Life Applications

Maxima and minima have many practical applications.

- **Business:** Used to maximize profit and minimize cost.
- **Engineering:** Used to design structures with maximum strength and minimum material.
- **Agriculture:** Used to obtain maximum crop production from available resources.

- **Transportation:** Used to minimize fuel consumption and travel cost.
- **Construction:** Used to maximize area and minimize the amount of material required.
- **Manufacturing:** Used to optimize production and reduce waste.

Importance of Maxima and Minima

The study of maxima and minima helps us make better decisions in practical situations.

It allows us to determine the best possible value of a quantity. By using derivatives, problems involving maximum profit, minimum cost, maximum area, minimum distance, and efficient resource usage can be solved mathematically.

6 APPLICATIONS OF DIFFERENTIATION IN ECONOMICS AND BUSINESS

Introduction

Differentiation is widely used in economics and business to study how different quantities change. It helps businesses understand the relationship between production, cost, revenue, and profit.

Derivatives are particularly useful for finding marginal cost, marginal revenue, marginal profit, and the production level that gives maximum profit.

Cost Function

The total cost of producing x units of a product can be represented by a cost function

$$C = C(x)$$

The total cost may include fixed costs and variable costs.

A simple cost function can be written as

$$C(x) = F + V(x)$$

where F represents fixed cost and $V(x)$ represents variable cost.

Marginal Cost

Marginal cost represents the rate at which total cost changes when the quantity of production changes.

It is given by

$$MC = \frac{dC}{dx}$$

Therefore, the derivative of the cost function gives the marginal cost.

Example:

Suppose the total cost is

$$C(x) = 500 + 20x + 2x^2$$

Then,

$$MC = \frac{dC}{dx}$$

Therefore,

$$MC = 20 + 4x$$

At $x = 10$,

$$MC = 20 + 4(10)$$

$$MC = 60$$

Therefore,

$$\boxed{MC = 60}$$

The marginal cost at 10 units of production is 60 cost units per additional unit.

Revenue Function

Revenue is the total amount of money received from selling a product.

If x units are sold at a price of p per unit, then

$$R = px$$

In general, revenue can be represented by

$$R = R(x)$$

where x is the number of units sold.

Marginal Revenue

Marginal revenue represents the rate of change of total revenue with respect to the number of units sold.

It is given by

$$MR = \frac{dR}{dx}$$

Therefore, differentiation can be used to determine how revenue changes when sales increase.

Example:

Suppose the revenue function is

$$R(x) = 100x - x^2$$

Then,

$$MR = \frac{dR}{dx}$$

Therefore,

$$MR = 100 - 2x$$

At $x = 20$,

$$MR = 100 - 2(20)$$

$$MR = 60$$

Hence,

$$\boxed{MR = 60}$$

Profit Function

Profit is the difference between total revenue and total cost.

Therefore,

$$P(x) = R(x) - C(x)$$

where

- $P(x)$ represents profit,
- $R(x)$ represents revenue,
- $C(x)$ represents cost.

The derivative of the profit function gives the marginal profit.

$$MP = \frac{dP}{dx}$$

Maximum Profit

Differentiation can be used to find the production level that gives maximum profit.

For maximum profit,

$$\frac{dP}{dx} = 0$$

After finding the critical point, the second derivative can be used to verify the maximum.

If

$$\frac{d^2P}{dx^2} < 0$$

then the profit is maximum.

Worked Example – Maximum Profit

Suppose the revenue and cost functions are

$$R(x) = 100x - x^2$$

and

$$C(x) = 20x + 100$$

The profit function is

$$P(x) = R(x) - C(x)$$

Therefore,

$$P(x) = 100x - x^2 - (20x + 100)$$

Simplifying,

$$P(x) = 80x - x^2 - 100$$

Differentiate,

$$\frac{dP}{dx} = 80 - 2x$$

For maximum profit,

$$80 - 2x = 0$$

Therefore,

$$2x = 80$$

Hence,

$$x = 40$$

Now find the second derivative.

$$\frac{d^2P}{dx^2} = -2$$

Since

$$\frac{d^2P}{dx^2} < 0$$

the profit is maximum at

$$\boxed{x = 40}$$

Therefore, the company should produce 40 units to obtain maximum profit.

Marginal Profit

Marginal profit is the rate of change of profit with respect to the number of units produced.

It is given by

$$MP = \frac{dP}{dx}$$

Since

$$P(x) = R(x) - C(x)$$

we get

$$\frac{dP}{dx} = \frac{dR}{dx} - \frac{dC}{dx}$$

Therefore,

$$\boxed{MP = MR - MC}$$

This relationship is useful for analysing business decisions.

Break-Even Point

A business reaches the break-even point when total revenue is equal to total cost.

Therefore,

$$R(x) = C(x)$$

At the break-even point,

$$P(x) = 0$$

because

$$P(x) = R(x) - C(x)$$

The break-even point is important because it shows the minimum sales level required to avoid a loss.

Worked Example – Break-Even Point

Suppose

$$R(x) = 50x$$

and

$$C(x) = 200 + 30x$$

At the break-even point,

$$R(x) = C(x)$$

Therefore,

$$50x = 200 + 30x$$

$$20x = 200$$

Hence,

$$\boxed{x = 10}$$

Therefore, the break-even point occurs when 10 units are sold.

Applications in Business

Differentiation is useful in many business decisions.

- **Profit Maximization:** Helps determine the production level that gives maximum profit.
- **Cost Minimization:** Helps businesses reduce production costs.
- **Revenue Analysis:** Helps study how revenue changes with sales.
- **Pricing Decisions:** Helps analyse the effect of changes in price on revenue.
- **Production Planning:** Helps determine efficient production levels.
- **Resource Management:** Helps businesses use available resources efficiently.

Importance in Economics

Differentiation provides a mathematical method for analysing changing economic quantities.

It is used to study marginal quantities such as marginal cost, marginal revenue, and marginal profit. It also helps economists and businesses make decisions related to production, pricing, and resource allocation.

7 OTHER REAL-LIFE APPLICATIONS OF DIFFERENTIATION

Introduction

Differentiation is not limited to mathematics and physics. It is also used in many practical areas such as engineering, medicine, population studies, environmental science, and technology.

Whenever a quantity changes with respect to another quantity, differentiation can be used to measure and analyse that change.

Population Growth

Population changes continuously with time. Differentiation can be used to determine the rate at which a population is increasing or decreasing.

Let the population be represented by

$$P = P(t)$$

where t represents time.

The rate of population change is

$$\frac{dP}{dt}$$

If

$$\frac{dP}{dt} > 0$$

the population is increasing.

If

$$\frac{dP}{dt} < 0$$

the population is decreasing.

Example:

Suppose the population of a town is

$$P(t) = 1000 + 50t + 2t^2$$

Then,

$$\frac{dP}{dt} = 50 + 4t$$

At $t = 5$,

$$\frac{dP}{dt} = 50 + 4(5)$$

$$= 70$$

Therefore, the population is increasing at a rate of

$$\boxed{70 \text{ people per unit time}}$$

Temperature Change

Temperature changes throughout the day. Differentiation can be used to determine how quickly the temperature changes with time.

Let the temperature be

$$T = T(t)$$

Then the rate of change of temperature is

$$\frac{dT}{dt}$$

A positive value indicates that the temperature is increasing, while a negative value indicates that the temperature is decreasing.

Example:

Suppose

$$T(t) = 20 + 4t - t^2$$

Then,

$$\frac{dT}{dt} = 4 - 2t$$

At $t = 1$,

$$\frac{dT}{dt} = 4 - 2(1)$$

$$= 2$$

Therefore, the temperature is increasing at a rate of

$$\boxed{2^{\circ}\text{C per unit time}}$$

Applications in Engineering

Differentiation is widely used in engineering to study changing quantities.

Engineers use derivatives to analyse motion, force, energy, temperature, pressure, and electrical signals.

For example, if the displacement of a machine part is represented by

$$s = s(t)$$

then its velocity is

$$v = \frac{ds}{dt}$$

and its acceleration is

$$a = \frac{d^2s}{dt^2}$$

Thus, differentiation helps engineers design and analyse mechanical systems.

Applications in Medicine

Differentiation can be used to study changes in biological quantities.

For example, the concentration of a medicine in the bloodstream may change with time. If the concentration is represented by

$$C = C(t)$$

then

$$\frac{dC}{dt}$$

represents the rate at which the concentration changes.

This type of mathematical analysis can help in understanding how quickly a substance enters or leaves the body.

Applications in Environmental Science

Environmental quantities such as pollution level, water level, temperature, and atmospheric conditions change with time.

Differentiation can be used to study these changes.

For example, if the amount of pollution in an area is represented by

$$P = P(t)$$

then

$$\frac{dP}{dt}$$

represents the rate at which the pollution level changes.

This can help researchers study environmental changes and develop suitable control methods.

Applications in Electricity

Differentiation is also important in electrical engineering.

If electric charge is represented by

$$Q = Q(t)$$

then electric current is the rate of change of charge with respect to time.

Therefore,

$$I = \frac{dQ}{dt}$$

where I represents electric current.

Thus, differentiation provides a mathematical relationship between electric charge and current.

Applications in Economics

Apart from cost and revenue, differentiation is also used to study demand and supply.

If the demand for a product depends on its price, it can be represented by

$$D = D(p)$$

The derivative

$$\frac{dD}{dp}$$

shows how the demand changes when the price changes.

This helps businesses and economists understand consumer behaviour.

Applications in Agriculture

Differentiation can be used in agriculture to study crop growth and resource usage.

Suppose crop production is represented by

$$P = P(x)$$

where x represents the amount of a particular resource such as water or fertilizer.

Then,

$$\frac{dP}{dx}$$

represents the rate at which crop production changes with respect to the resource used.

This can help farmers use resources efficiently and improve production.

Applications in Computer Science

Differentiation is also used in computer science and artificial intelligence.

Many algorithms use derivatives to find the minimum value of an error function. This helps a computer system improve its performance.

For example, in machine learning, an error function can be represented by

$$E = E(x)$$

The derivative

$$\frac{dE}{dx}$$

helps determine the direction in which the error should be reduced.

Therefore, differentiation is an important mathematical concept in modern technology.

Applications in Architecture and Construction

Differentiation can help architects and engineers optimize the design of buildings and structures.

It can be used to determine maximum area, minimum material usage, suitable dimensions, and efficient structural designs.

For example, optimization methods based on differentiation can help find dimensions that provide maximum space while using a limited amount of construction material.

Applications in Sports

Differentiation is useful for analysing the movement of athletes and sports equipment.

The position of a player or ball can be represented as a function of time.

The velocity is

$$v = \frac{ds}{dt}$$

and acceleration is

$$a = \frac{d^2s}{dt^2}$$

This can be used to study running speed, ball movement, jumping motion, and other sports activities.

Importance of Differentiation in Real Life

Differentiation helps us understand how quantities change. It provides a mathematical method for measuring these changes and making useful predictions.

Its applications can be found in

- Physics
- Engineering
- Economics
- Business
- Medicine
- Agriculture
- Environmental Science
- Computer Science
- Sports
- Construction

8 WORKED EXAMPLES

Introduction

Differentiation can be applied to many practical problems. The following worked examples show how derivatives can be used to solve problems related to motion, maxima and minima, business, and rate of change.

Each example is solved step by step to make the application of differentiation easy to understand.

Example – Finding Velocity

The position of a particle is given by

$$s = t^3 + 4t^2 + 2t$$

Find the velocity at $t = 2$ seconds.

Velocity is the rate of change of position with respect to time.

Therefore,

$$v = \frac{ds}{dt}$$

Differentiating,

$$v = 3t^2 + 8t + 2$$

At $t = 2$,

$$v = 3(2)^2 + 8(2) + 2$$

$$= 12 + 16 + 2$$

Therefore,

$$\boxed{v = 30 \text{ m/s}}$$

Hence, the velocity of the particle at $t = 2$ seconds is 30 m/s.

Example – Finding Acceleration

The velocity of a particle is given by

$$v = 4t^2 + 6t + 2$$

Find the acceleration at $t = 3$ seconds.
Acceleration is the rate of change of velocity.
Therefore,

$$a = \frac{dv}{dt}$$

Differentiating,

$$a = 8t + 6$$

At $t = 3$,

$$a = 8(3) + 6$$

$$= 24 + 6$$

Therefore,

$$\boxed{a = 30 \text{ m/s}^2}$$

Hence, the acceleration at $t = 3$ seconds is 30 m/s^2.

Example – Maximum Area

A rectangular field has a fixed perimeter of 40 metres. Find the dimensions that give the maximum area.
Let the length be x metres and the width be y metres.
The perimeter is

$$2x + 2y = 40$$

Therefore,

$$x + y = 20$$

Hence,

$$y = 20 - x$$

The area is

$$A = xy$$

Therefore,

$$A = x(20 - x)$$

Simplifying,

$$A = 20x - x^2$$

Differentiate,

$$\frac{dA}{dx} = 20 - 2x$$

For maximum area,

$$\frac{dA}{dx} = 0$$

Therefore,

$$20 - 2x = 0$$

$$2x = 20$$

Hence,

$$x = 10$$

Since

$$y = 20 - x$$

we get

$$y = 10$$

Therefore, the dimensions are

$$\boxed{x = 10 \text{ m}, \qquad y = 10 \text{ m}}$$

The maximum area is

$$A = 10 \times 10$$

Therefore,

$$\boxed{A = 100 \text{ m}^2}$$

Hence, the maximum area is obtained when the rectangular field becomes a square.

Example – Maximum Profit

Suppose the revenue of a company is

$$R(x) = 80x - x^2$$

and the cost is

$$C(x) = 20x + 100$$

Find the production level that gives maximum profit.
The profit function is

$$P(x) = R(x) - C(x)$$

Therefore,

$$P(x) = 80x - x^2 - (20x + 100)$$

Simplifying,

$$P(x) = 60x - x^2 - 100$$

Differentiate,

$$\frac{dP}{dx} = 60 - 2x$$

For maximum profit,

$$\frac{dP}{dx} = 0$$

Therefore,

$$60 - 2x = 0$$

$$2x = 60$$

Hence,

$$x = 30$$

Now find the second derivative.

$$\frac{d^2P}{dx^2} = -2$$

Since

$$\frac{d^2P}{dx^2} < 0$$

the profit is maximum.
Therefore, the production level for maximum profit is

$$\boxed{x = 30}$$

The maximum profit is

$$P(30) = 60(30) - (30)^2 - 100$$

$$= 1800 - 900 - 100$$

Therefore,

$$\boxed{P(30) = 800}$$

Hence, the maximum profit is 800 monetary units.

Example – Rate of Population Growth

The population of a town is given by

$$P(t) = 2000 + 100t + 5t^2$$

Find the rate of population growth at $t = 4$.
The rate of population growth is

$$\frac{dP}{dt}$$

Differentiating,

$$\frac{dP}{dt} = 100 + 10t$$

At $t = 4$,

$$\frac{dP}{dt} = 100 + 10(4)$$

$$= 100 + 40$$

Therefore,

$$\boxed{\frac{dP}{dt} = 140}$$

Hence, the population is increasing at a rate of 140 people per unit time.

Example – Minimum Cost

Suppose the cost of producing x units is

$$C(x) = x^2 - 12x + 50$$

Find the production level at which the cost is minimum.
Differentiate the cost function.

$$\frac{dC}{dx} = 2x - 12$$

For minimum cost,

$$\frac{dC}{dx} = 0$$

Therefore,

$$2x - 12 = 0$$

$$2x = 12$$

Hence,

$$x = 6$$

Now find the second derivative.

$$\frac{d^2C}{dx^2} = 2$$

Since

$$\frac{d^2C}{dx^2} > 0$$

the cost is minimum at $x = 6$.
The minimum cost is

$$C(6) = 6^2 - 12(6) + 50$$

$$= 36 - 72 + 50$$

Therefore,

$$\boxed{C(6) = 14}$$

Hence, the minimum cost is 14 monetary units.

Example – Chain Rule

Find the derivative of

$$y = (2x^2 + 3)^4$$

Using the chain rule,

$$\frac{dy}{dx} = 4(2x^2 + 3)^3(4x)$$

Therefore,

$$\boxed{\frac{dy}{dx} = 16x(2x^2 + 3)^3}$$

Example – Product Rule

Find the derivative of

$$y = x^2(x+5)$$

Using the product rule,

$$\frac{dy}{dx} = x^2\frac{d}{dx}(x+5) + (x+5)\frac{d}{dx}(x^2)$$

Therefore,

$$\frac{dy}{dx} = x^2(1) + (x+5)(2x)$$

Simplifying,

$$\frac{dy}{dx} = x^2 + 2x^2 + 10x$$

Hence,

$$\boxed{\frac{dy}{dx} = 3x^2 + 10x}$$

Example – Quotient Rule

Find the derivative of

$$y = \frac{x^2+1}{x+1}$$

Using the quotient rule,

$$\frac{dy}{dx} = \frac{(x+1)(2x) - (x^2+1)(1)}{(x+1)^2}$$

Simplifying,

$$\frac{dy}{dx} = \frac{2x^2 + 2x - x^2 - 1}{(x+1)^2}$$

Therefore,

$$\boxed{\frac{dy}{dx} = \frac{x^2+2x-1}{(x+1)^2}}$$

9 ADVANTAGES AND IMPORTANCE OF DIFFERENTIATION

Introduction

Differentiation is one of the most important concepts in calculus. It provides a mathematical method for studying how one quantity changes with respect to another.

It has applications in mathematics, science, engineering, economics, business, medicine, and many other fields.

Understanding Rate of Change

One of the main advantages of differentiation is that it helps us understand the rate of change of a quantity.

If

$$y = f(x)$$

then

$$\frac{dy}{dx}$$

represents the rate at which y changes with respect to x.

This helps us study continuously changing quantities in real-life situations.

Studying Motion

Differentiation is very useful in the study of motion.

If s represents position, then velocity is given by

$$v = \frac{ds}{dt}$$

and acceleration is given by

$$a = \frac{d^2s}{dt^2}$$

Therefore, differentiation helps us understand the movement of cars, trains, aircraft, sports objects, and other moving bodies.

Finding Maximum and Minimum Values

Differentiation helps to find the maximum and minimum values of functions.

For a critical point,

$$\frac{dy}{dx} = 0$$

The second derivative can then be used to determine whether the point represents a maximum or minimum.

This is useful in optimization problems such as maximum profit, minimum cost, maximum area, and minimum material usage.

Applications in Business

Differentiation helps businesses make better decisions.

It can be used to determine

- Maximum profit
- Minimum cost
- Marginal cost
- Marginal revenue
- Marginal profit
- Optimum production level

Thus, differentiation provides useful information for business planning and decision-making.

Applications in Engineering

Engineers use differentiation to study and design mechanical, electrical, and structural systems.

It is used to analyse motion, force, energy, temperature, pressure, electrical signals, and many other quantities.

Differentiation also helps engineers optimize designs and improve the efficiency of machines and structures.

Applications in Science

Differentiation is widely used in physics, chemistry, biology, and environmental science.

It helps scientists study changing quantities such as velocity, acceleration, temperature, population, chemical concentration, and environmental conditions.

Therefore, differentiation provides an important mathematical foundation for scientific analysis.

Applications in Economics

In economics, differentiation is used to study marginal quantities.

For example, if $C(x)$ represents the total cost, then marginal cost is

$$MC = \frac{dC}{dx}$$

Similarly, if $R(x)$ represents total revenue, then marginal revenue is

$$MR = \frac{dR}{dx}$$

These concepts help economists and businesses analyse production and pricing decisions.

Applications in Medicine

Differentiation can be used to study how medical and biological quantities change with time.

For example, if $C(t)$ represents the concentration of a medicine in the bloodstream, then

$$\frac{dC}{dt}$$

represents the rate at which the concentration changes.

This type of analysis can help in understanding the behaviour of substances inside the body.

Applications in Technology

Modern technology also makes use of differentiation.

In computer science and machine learning, derivatives are used to optimize mathematical functions and reduce errors.

For example, if an error function is represented by

$$E = E(x)$$

then

$$\frac{dE}{dx}$$

can be used to determine how the error changes and how it can be reduced.

Improved Decision Making

Differentiation helps us identify the best possible solution among different choices.

For example, a company can use differentiation to determine the production level that gives maximum profit. Similarly, an engineer can use optimization to find a design that uses minimum material.

Therefore, differentiation supports effective decision-making.

Mathematical Modelling

Differentiation is an important part of mathematical modelling.

Real-life situations can be represented using mathematical functions. Differentiation can then be applied to these functions to understand how the system changes.

This allows complex real-life problems to be studied using mathematical methods.

Major Advantages

The major advantages of differentiation can be summarized as follows:

- It measures the rate of change.
- It helps to study motion.
- It helps to find maximum and minimum values.
- It is useful for optimization.
- It helps businesses maximize profit and minimize cost.
- It is widely used in engineering and science.
- It helps analyse changing quantities.
- It supports mathematical modelling.
- It helps in making practical decisions.
- It has applications in modern technology.

10 CONCLUSION

Introduction

Differentiation is one of the most important concepts in mathematics. It helps us understand how one quantity changes with respect to another. Although differentiation is a mathematical technique, it has many practical applications in real life.

In this project, we studied the basic concepts of differentiation and its applications in different fields such as motion, business, economics, science, engineering, medicine, technology, and population studies.

Importance of Differentiation

Differentiation provides a mathematical method to measure the rate of change of a quantity. It is useful for studying both small and instantaneous changes.

For example, if the position of an object is given by $s(t)$, its velocity can be found using

$$v = \frac{ds}{dt}$$

and its acceleration is given by

$$a = \frac{d^2s}{dt^2}.$$

Thus, differentiation helps us understand the motion of objects accurately.

Applications in Real Life

Differentiation is widely used in many real-life situations.

- In physics, it is used to find velocity, acceleration, and rates of change.
- In economics, it is used to calculate marginal cost, marginal revenue, and marginal profit.
- In business, it helps to maximize profit and minimize cost.
- In engineering, it is used to study changes in physical quantities and optimize designs.
- In medicine, it can be used to study changes in drug concentration and patient-related measurements.
- In population studies, differentiation helps to measure population growth.
- In environmental science, it helps to study changes in temperature, pollution, and other environmental factors.
- In technology and computer science, differentiation is used in optimization and machine learning.

Maximum and Minimum

One of the important applications of differentiation is finding maximum and minimum values.

If

$$\frac{dy}{dx} = 0,$$

then the corresponding point may be a maximum or minimum point. The second derivative helps to identify the nature of the point.

For a maximum,

$$\frac{d^2y}{dx^2} < 0,$$

and for a minimum,

$$\frac{d^2y}{dx^2} > 0.$$

This concept is very useful in business and economics for maximizing profit and minimizing cost.

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Online Resources

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4. MIT OpenCourseWare, Single Variable Calculus.

Topics Referred

The following mathematical concepts were referred to while preparing this project:

- Basic rules of differentiation
- Rate of change
- Velocity and acceleration
- Maxima and minima
- Cost, revenue and profit functions
- Marginal cost and marginal revenue
- Population growth
- Optimization problems
- Applications of differentiation in science and engineering